

Why Learn AI?

Pretend you're a scribe in the 1450s. All day, you hand-copy the king's proclamations and send them out across the kingdom. Good work. Steady job. Then one morning, you hear that a guy in Mainz built a machine that prints pages a thousand times faster than you can write them. Two choices: pretend it isn't happening, or learn the new machine well enough to run it.

AI Is the Press



✓ Run it, or someone else will.

AI IS EVERYWHERE

AI isn't something you go visit. It's already in the apps on your phone, the search results you read, and the tools your first job will hand you on day one. It feels like AI went from sci-fi to normal in a few years. **Most of what it does today, it will do better tomorrow.**

Where AI Already Lives

Recommends

Predicts what you might like next.

YOU'VE SEEN IT IN

Spotify · Netflix
TikTok

Navigation

Predicts traffic and arrival time.

YOU'VE SEEN IT IN

Google Maps
Waze

Face Recognition

Checks whether a face matches you.

YOU'VE SEEN IT IN

Phone unlock
Photo tagging

Voice Assistants

Turns your speech into words.

YOU'VE SEEN IT IN

Siri · Alexa
Hey Google

Chatbots

Carries on a conversation with you.

YOU'VE SEEN IT IN

ChatGPT · Claude
Gemini

✓ AI was already part of your day before chatbots arrived.

YOU CAN START NOW

Here's the great news: you don't have to wait until college or your first job to get good at this. You can start now. Learn what AI does well, notice where it struggles, and practice using it for something you care about. Every project gives you experience you can bring to the next one.

Before personal computers, becoming a designer meant years at a drafting table learning design by hand. Then desktop publishing put powerful design tools on a teenager's desk. They could create posters, magazines, and brochures while developing their skills. The tool didn't replace skill. It shortened the distance between wanting to do the work and actually doing it.

Why You'll Thrive in the AI Future



This Is Your Time

Start building AI skills while you're in school. Try new things, ask questions, and learn from what happens.



You'll Move Faster

Try ideas and get feedback on your work. AI can help you tackle projects you wouldn't have known how to start. Learn by building, starting now.



Build Good Habits Early

Use AI to learn and work from the start. Practice asking good questions, checking answers, and making your own decisions. These habits will stay with you.

✓ Start now. Build skills you'll carry into whatever comes next.

THIS HAS HAPPENED BEFORE

AI isn't the first technology to transform how people work. The steam engine did it for physical labor, electricity for factories, and the internet for information. **AI could do it across almost everything.**

In July 2025, the White House released an official national strategy document, *Winning the Race: America's AI Action Plan*. Here's how it describes what AI makes possible:

“

An industrial revolution, an information revolution, and a renaissance—all at once. This is the potential that AI presents.

— WINNING THE RACE · The White House, July 2025

AI is today's printing press.

Learn to run it.

What Is AI?

Ask your desk at school, “Give me 10 ideas for my next history project.”

Nothing. It’s a desk.

Ask the Desk. Ask AI.



✓ AI is software built to do things that used to take a human brain.

Ask a friend, and they can help you brainstorm. Ask AI, and it can give you a list in seconds.

AI is software built to do things that used to take a human brain. Understand a question. Summarize information. Translate a conversation. Suggest ideas for a history project.

That doesn’t mean it thinks like your friend, or that all 10 ideas will be good. But you now have a tool that can help with tasks that once needed a person.

Different AI systems do different jobs. Let’s look at two kinds you already use.

Two Ways You Already Use AI



Recommendation AI

THE JOB

Choose from what already exists.

HOW IT WORKS

Rank the available options and select the likely best match.

EVERYDAY EXAMPLES

Your next Netflix show, Spotify song, or Maps route.



Generative AI

THE JOB

Make something that didn’t exist.

HOW IT WORKS

Use learned patterns to create a new output from your prompt (the question or instructions you give it).

EVERYDAY EXAMPLES

An email, essay, image, website, song, or video.

✓ AI can recommend. AI can create. This course focuses on generative AI.

Here’s what that difference looks like.

One Picks. One Creates.

THE SCENARIO

You’re in the mood for superheroes.



PICKS

Recommendation AI

Chooses from what already exists

THE JOB

Find a superhero movie you might enjoy.

WHAT YOU GET

Captain America: Civil War

A movie selected from an existing catalog based on your interests.

It picked a movie that already exists.



CREATES

Generative AI

Creates something new from your request

THE JOB

Write a scene about two superhero teammates who disagree.

WHAT YOU GET

“We save the bridge,” Maya said. “The hospital loses power in three minutes,” Leo replied. “We can’t do both.”

It generated a scene from your request.

✓ One helps you find something to watch. The other helps you create a story of your own.

Two kinds. One picks, one creates.

This course is about the one that creates.

How an LLM Works

If you've used AI, you've probably used an app like ChatGPT, Claude, or Gemini. The app is what you use. Under the hood, a **Large Language Model**, or **LLM** for short, does the work.

What's an LLM?



Large

Trained on huge amounts of text and code.



Language

Reads, writes, summarizes, translates, and explains.



Model

Predicts likely output from learned patterns.

✓ ChatGPT is the app. The LLM is the engine.

So how does an LLM actually turn your words into an answer?

When ChatGPT or Claude write a sentence, they're running math to predict the likely next words. They aren't looking up what your words mean; they're working out which words tend to follow which.

How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.

Learn Once. Answer Every Word.

LEARN ONCE

01

Training

The model learns from enormous amounts of data once, before you use it.

02

Patterns

Training turns examples into learned numerical patterns.

Patterns power every answer

ANSWER - EVERY WORD

03

Probability

For every next word, the model scores what is most likely.

04

Prediction

It chooses one likely next word, then runs the process again.

✓ Learn once. Use the patterns for every answer.

Four ideas carry it. Below, we follow one example, peanut butter, through all four.

01 Training

The model **teaches itself**: guess the next word, check, and nudge its numbers toward the right word.

How Training Works



1

Read

The model reads a training example with the answer included.



2

Guess

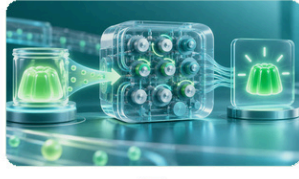
Given 'Peanut butter and ___,' the model guesses cloud.



3

Check

The example says jelly. Compare the guess with that word.



4

Adjust

Adjust the model's internal numbers to make jelly more likely in this situation.

NEXT EXAMPLE

✓ Repeat with more examples. The patterns build.

02 Patterns

So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?

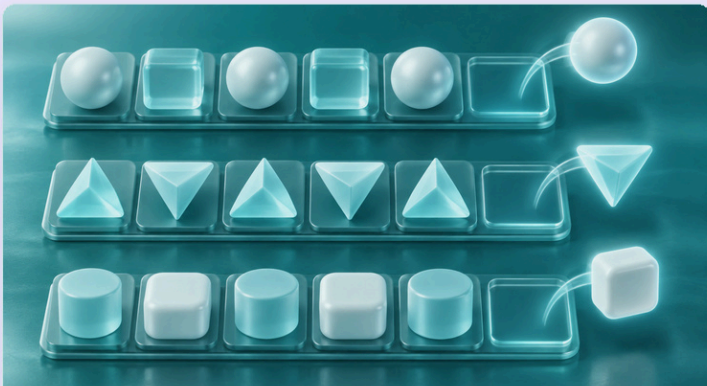
How AI Learns Patterns



One Familiar Pattern

Peanut butter and ___ → **jelly**

You knew it. So does AI.



Patterns Are Everywhere

Twinkle, twinkle, little ___ → **star**

Once upon a ___ → **time**

Better late than ___ → **never**

✓ AI learns patterns by working through billions of examples.

These are easy patterns you already know. AI also learns patterns in places you might not expect: how people explain ideas, ask questions, solve problems, and even misspell words.

03 Probability

AI doesn't make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.

Same Word. Different Odds.

I'd like to buy peanut butter and _____.

jelly		41%
bread		27%
bananas		16%
honey		5%

I'd like to buy a peanut butter and banana _____.

sandwich		54%
smoothie		16%
toast		9%
jelly		2%

✓ The surrounding words change the odds.

In this example, adding "banana" drops the probability of "jelly" from 41% to 2%.

04 Prediction

Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next.

One Word at a Time

I want to buy peanut butter and

→ **jelly**



I want to buy peanut butter and

jelly

→ **for**



I want to buy peanut butter and

jelly for

→ **lunch**

✓ Add a word. Use the updated sentence. Predict again.

A full paragraph runs this loop many times, fast enough to look like thought.

Training builds the patterns.

The model uses those patterns to build your answer, one word at a time.

Does AI Think?

You type a question and AI types back like a person. It explains, it jokes, it says sorry when it slips up. After a while it’s hard not to feel like someone is in there. So here’s the big question: **Does AI actually think?** When it answers like a person, is it understanding things the way you do?

SOUNDING HUMAN

AI can finish your sentence, explain a poem, and carry on a conversation. But sounding like a person doesn’t tell you whether it understands things the way a person does.

There’s a famous way to picture this, sometimes called the Chinese Room. Imagine someone locked in a room who doesn’t know a word of Chinese. Notes in Chinese come in under the door. The person can’t read any of it, but they have a giant rulebook telling them which symbols to send back. So they match symbols and slide back an answer that looks perfect to the people outside.

The Chinese Room

1 People outside slide a note in Chinese under the door.

2 Inside, the person can't read it. They look up the whole phrase on a giant chart.

3 They follow the rulebook, choose the matching reply, and slide it back under the door.

To anyone outside, it looks like fluent Chinese conversation.

你好嗎? (How are you?)

我很好, 謝謝! (I'm fine, thank you!)

Rules Only

我不懂中文

中文句子 → 回覆對照表

如果你看到:	請回覆:
你好嗎? →	我很好, 謝謝!

所有中文句子與回覆

1. 查看句子 ✓
2. 找到對應回覆 ✓
3. 拿出正確紙條 ✓
4. 從門縫滑回去 ✓

✓ A convincing answer doesn’t prove understanding.

The Chinese Room illustrates a distinction: producing an answer and understanding it aren’t necessarily the same thing. An LLM doesn’t follow a literal rulebook. It uses patterns learned during training to predict its response. A fluent answer alone doesn’t tell you whether it understands.

When You Think. What AI Does.

When You Think	What AI Does
Meaning. You connect words to what they mean in your life.	AI uses learned patterns and what you tell it.
Experience. You draw on things you have lived through.	AI works with patterns learned from text, images, audio, and other data.
Word choice. You choose words to express what you want to say.	AI predicts the next word, then repeats.
Beauty. You have a personal response to art or music.	AI can describe beauty using learned patterns.
Uncertainty. You can notice when you are unsure.	AI can sound certain even when it is wrong.

✓ Similar-looking answers can come from very different processes.

None of this means AI is dumb or useless. It can do impressive work. But an answer that sounds human is not enough to show that it understands the way you do.

Sounds human. Works differently.

A convincing answer doesn’t prove understanding.

What You Can Control

You’re hearing it constantly: AI is taking jobs. That headline used to be a prediction. Now it’s news.

And jobs are just the loudest headline. AI is also reshaping who holds power, what it costs the planet, and what you can trust online. Which way it all goes, nobody knows yet, including the people making the predictions. And almost none of it is in your hands.

So here’s the question worth asking: what’s in your hands, and what isn’t?

What’s in Your Hands?



Out of Your Hands	In Your Hands
How AI changes future jobs	Build real depth in something you care about.
How quickly AI changes	Stay curious and keep what works.
How much work AI can produce	Practice coming up with your own ideas.
How other people use AI	Choose how you use AI and how you work with other people.
What the next headline predicts	Make something real instead of waiting to see what happens.

✓

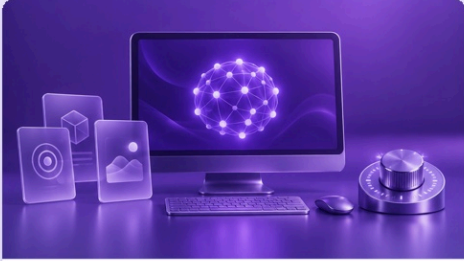
You don’t control every outcome. Your choices still matter.

Keep informed about the left column. Put most of your energy into the right.

SO WHAT DO YOU DO?

That right column isn’t a feeling, it’s a to-do list. Three moves worth your energy:

Three Moves Worth Your Energy



Go Deep

Pick one AI tool and learn what it does well, where it gets things wrong, and how to push it. Depth beats dabbling.

Think First

Form your own take before you ask. Then use AI to sharpen your thinking. That’s the difference between getting smarter and just getting answers.

Skip the Hype

You can’t control the headlines. You can control whether you doomscroll them or spend that hour getting better at something real.

✓

Put your effort where it changes your outcome.

You can’t control where AI goes next.

You can build the skill and judgment to decide what you do next.

Does School Matter?

A lot of people your age are quietly asking the same thing. If AI can answer anything, do you even need school anymore? Why sit through years of classes when the tool spits out the answer in seconds?

Think about it this way.

Fast forward a few years. You’ve landed your dream job, and AI sits right next to you all day, cranking out drafts, code, and plans. Your coworker at the next desk is doing the same. You each have the same job, using the same AI.

Same Tool. Different Advantage.



✓ The tool may be the same. What you bring to it is yours.

Here’s the catch: ask AI similar questions, and you often get similar answers. They might be good. They might need work. Either way, that starting point is available to everyone. It’s the new average. What you bring to the work is what takes you further.

That leads to a question worth considering: In your AI future, what takes you beyond the new average? What makes you more valuable?

WHAT SETS YOU APART

Your secret sauce is the knowledge and skills you build. And where do you build them? You know where this is going: school.

School is where you get to build what you bring to the work. Learn how things work. Practice writing, solving problems, making things, and working with people. Find something that interests you and go deep. Whatever tools come next, that knowledge and those skills are yours. That’s what it means to **Be Smarter Than the Tool**.

What to Start Building Today



Deep Subject Knowledge
Go deep in a subject you care about. Become someone people turn to for your knowledge.



Strong Skills
Practice writing, building, analyzing, and solving problems. Get good at turning what you know into work you’re proud of.



AI Fluency
Know how to work with AI, challenge it, and use it to go beyond the average.



People Skills
Listen, communicate, earn trust, and help a team turn ideas into action.

✓ School helps you build what takes you beyond the new average.

The opportunity to learn is already in front of you.

Use it to build the knowledge and skills that take you beyond the new average.

Learn with AI

Do you ever feel like there aren't enough hours in the day? You might have a history quiz tomorrow, an essay due Friday, and what feels like a never-ending stream of algebra homework. Add sports, activities, family, and everything else, and finding enough time to study feels impossible.

There's an easy solution: add four more hours to every day. Then you'd have plenty of time.

Of course, that isn't possible. AI can't add hours to your day, either.

THE PATIENT TUTOR

Here's the good news: AI might be the best study tool you'll ever get your hands on. Used the right way, it helps you finish the work and makes you genuinely sharper at the subject.

Believe it or not, this is one place where a machine has an advantage over a human tutor. AI will quiz you at 1 a.m., never sigh when you ask again, and meet you exactly where you're stuck.

The first move is choosing which kind of AI to aim at the job. There's a simple question to guide your choice: do you want to learn from materials you already have, or learn something new?

Which Study Tool for the Job?



GEMINI NOTEBOOK

Focus

Uses your materials to answer questions and create study tools. That keeps your exam prep focused on what you need to learn.

BEST USE

You have the materials the test covers, including class notes, screenshots, study guides, a webpage, or a YouTube video.

THE CATCH

If your notes are missing a key concept, it cannot reliably fill the gap.



CHATGPT · CLAUDE · GEMINI

Exploration

Makes things from scratch. Trained on massive amounts of data, it can explain or quiz you on almost anything.

BEST USE

You do not have the materials, or you want a concept explained a new way or extra practice your notes do not cover.

THE CATCH

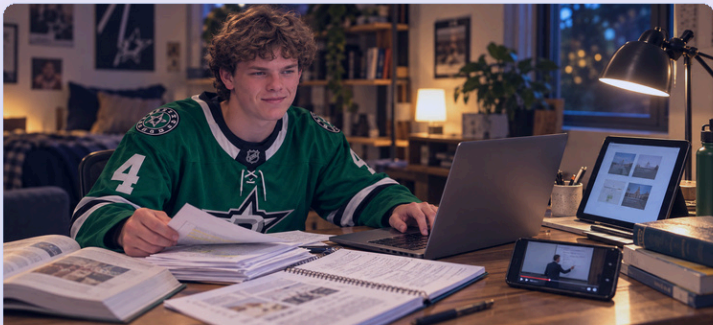
It can make things up or explain a concept differently from how your teacher taught it and expects it on the exam.

✓ Choose the tool that matches how you need to learn.

WHY GEMINI NOTEBOOK

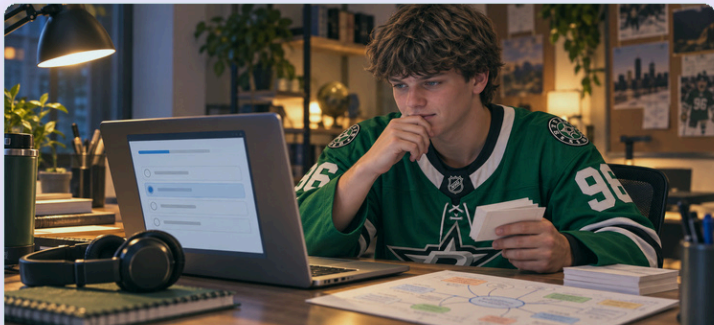
Use Gemini Notebook as your main study tool when you're studying from class materials. It keeps your work grounded in the notes, slides, and sources your teacher gave you.

How Gemini Notebook Works



You Upload

Your notes and PDFs, slides, images, websites and articles, YouTube videos, and more.



You Get

Quizzes, flashcards, study guides, audio and video overviews, mind maps, and more.

✓ Turn your class materials into the study tools you need.

Your Four Moves for Gemini Notebook



One Subject per Notebook

Mixing subjects is like asking, "Quiz me on history, but make some of the questions about math."



Give It the Full Picture

Add your notes, slides, videos, and anything else you need to learn.



Quiz Yourself Blind

Tell Gemini Notebook: "Generate a 10-question quiz from these sources only. Don't show the answers yet. Wait for mine, then grade me."



Trace It Back

If something is unclear, click the citation. It takes you back to the original material.

✓ Use AI to strengthen the learning, not skip it.

Use AI to learn, not to skip the learning.

Feed it your materials. Trace it back.